CLASS 8 · CBSE · GANITA PRAKASH PART 1 · DPP SHEET

Chapter 5: Number Play

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Is $8a - 6b + 4$ always even for integers a and b? Give a reason.
2. The sum of three consecutive numbers is 96. Find them.
3. Write an expression for all numbers that leave remainder 4 on division by 7, and list the first four.
4. Two numbers leave remainders 5 and 6 on division by 8. What remainder does their sum leave? Their product?
5. Without dividing, find the remainder when 86435 is divided by 9.
6. Test 72952 for divisibility by 11.
7. $5x28$ is divisible by 12. Find the digit x.
8. Find the digital roots of 9875 and of $9875 + 18$.
9. Always, sometimes or never true: "If a number is divisible by 9 and 6, it is divisible by 54."
10. Solve the cryptarithm ON + ON + ON = PO.

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Answer Key

1. Yes; it equals $2(4a - 3b + 2)$
2. 31, 32, 33
3. $7k + 4$: 4, 11, 18, 25
4. Sum: 3. Product: 6.
5. Digit sum 26, then 8; remainder 8
6. $2 - 5 + 9 - 2 + 7 = 11$, so it is divisible by 11
7. 28 is divisible by 4; $15 + x$ must be a multiple of 3, so $x = 0, 3, 6$ or 9
8. 2 and 2
9. Sometimes true (18 is not; 54 is). It is always divisible by 18.
10. $31 + 31 + 31 = 93$

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